2.17 Hyperbolic Functions: Reciprocal Functions

Worked Solutions

1 a)

$$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}} = \frac{2e^x}{e^{2x} - 1}.$$

1 b)

$$\operatorname{sech} x = \frac{2}{e^x + e^{-x}} = \frac{2e^x}{e^{2x} + 1}.$$

1 c)

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{e^{2x} + 1}{e^{2x} - 1}.$$

2 a)

$$\operatorname{cosech} 1 = \frac{2}{e - e^{-1}} = 0.851 \text{ (3 SF)}.$$

2 b)

$$\operatorname{sech} 1 = \frac{2}{e + e^{-1}} = 0.648 \text{ (3 SF)}.$$

2 c)

$$\operatorname{cosech}(-1) = \frac{2}{e^{-1} - e} = -0.851 \text{ (3 SF)}.$$

2 d)

$$\operatorname{sech}(-1) = \frac{2}{e^{-1} + e} = 0.648 \text{ (3 SF)}.$$

3)

$$\begin{aligned}
\operatorname{cosech}(-x) &\equiv \frac{2}{e^{-x} - e^{-(-x)}} \\
&\equiv \frac{2}{e^{-x} - e^{x}} \\
&\equiv \frac{2}{-(e^{x} - e^{-x})} \\
&\equiv -\frac{2}{e^{x} - e^{-x}} \\
&\equiv -\operatorname{cosech} x.
\end{aligned}$$

4)

$$\begin{aligned}\operatorname{sech}(-x) &\equiv \frac{2}{e^{-x}+e^{-(-x)}}\\ &\equiv \frac{2}{e^{-x}+e^{x}}\\ &\equiv \frac{2}{e^{x}+e^{-x}}\\ &\equiv \operatorname{sech} x.\end{aligned}$$

5 a)

$$\begin{aligned}\operatorname{cosech}(\ln 3) &\equiv \frac{2}{e^{\ln 3}-e^{-\ln 3}}\\ &\equiv \frac{2}{e^{\ln 3}-e^{\ln(1/3)}}\\ &\equiv \frac{2}{3-\frac{1}{3}} = \frac{3}{4}.\end{aligned}$$

5 b)

$$\begin{aligned}\operatorname{sech}(\ln 6) &\equiv \frac{2}{e^{\ln 6}+e^{-\ln 6}}\\ &\equiv \frac{2}{6+\frac{1}{6}} = \frac{12}{37}.\end{aligned}$$

5 c)

$$\begin{aligned}\operatorname{cosech}(\ln\sqrt{5}) &\equiv \frac{2}{e^{\ln\sqrt{5}}-e^{-\ln\sqrt{5}}}\\ &\equiv \frac{2}{\sqrt{5}-\frac{1}{\sqrt{5}}} = \frac{2\sqrt{5}}{5-1} = \frac{\sqrt{5}}{2}.\end{aligned}$$

5 d)

$$\begin{aligned}\operatorname{sech}(4\ln 3) &\equiv \frac{2}{e^{4\ln 3}+e^{-4\ln 3}}\\ &\equiv \frac{2}{e^{\ln 81}+e^{\ln(1/81)}}\\ &\equiv \frac{2}{81+\frac{1}{81}} = \frac{81}{3281}.\end{aligned}$$

6)

$$\begin{aligned}\operatorname{cosech}(\ln k) &= \frac{2}{e^{\ln k} - e^{-\ln k}} \\ &= \frac{2}{e^{\ln k} - e^{\ln(1/k)}} = \frac{2}{k - 1/k} \\ &= \frac{2k}{k^2 - 1}.\end{aligned}$$

7)

$$\begin{aligned}\operatorname{sech}(3\ln k) &= \frac{2}{e^{3\ln k} + e^{-3\ln k}} \\ &= \frac{2}{e^{\ln k^3} + e^{\ln(1/k^3)}} = \frac{2}{k^3 + 1/k^3} \\ &= \frac{2k^3}{k^6 + 1}.\end{aligned}$$

8 a)

$$\begin{aligned}f(1) &= \operatorname{cosech}(2\ln 1) - \operatorname{sech}(2\ln 1) \\ &= \operatorname{cosech} 0 - \operatorname{sech} 0 = \frac{1}{\sinh 0} - \frac{1}{\cosh 0} = \frac{1}{0} - \frac{1}{1}.\end{aligned}$$

So $f(x)$ is undefined when $x = 1$, hence $k = 1$.

8 b)

$$\begin{aligned}f(x) &= \operatorname{cosech}(2\ln x) - \operatorname{sech}(2\ln x) \\ &= \frac{2}{e^{2\ln x} - e^{-2\ln x}} - \frac{2}{e^{2\ln x} + e^{-2\ln x}} \\ &= \frac{2}{x^2 - x^{-2}} - \frac{2}{x^2 + x^{-2}} \\ &= \frac{2x^2}{x^4 - 1} - \frac{2x^2}{x^4 + 1} \\ &= \frac{2x^2(x^4+1) - 2x^2(x^4-1)}{(x^4-1)(x^4+1)} \\ &= \frac{2x^6 + 2x^2 - 2x^6 + 2x^2}{(x^4-1)(x^4+1)} \\ &= \frac{4x^2}{(x^4-1)(x^4+1)}.\end{aligned}$$

9)

$$\coth x \equiv \frac{\cosh x}{\sinh x}$$

$$\equiv \frac{(e^x + e^{-x})/2}{(e^x - e^{-x})/2}$$
$$\equiv \frac{e^x + e^{-x}}{e^x - e^{-x}}$$
$$\equiv \frac{e^{2x} + 1}{e^{2x} - 1}$$
$$\equiv \frac{e^{2x} + 1}{(e^x + 1)(e^x - 1)}.$$

10 a)

$$\coth(\ln 2) \equiv \frac{e^{2\ln 2} + 1}{e^{2\ln 2} - 1} = \frac{e^{\ln 4} + 1}{e^{\ln 4} - 1}$$
$$\equiv \frac{4 + 1}{4 - 1} = \frac{5}{3}.$$

10 b)

$$\coth(\ln \sqrt{2}) \equiv \frac{e^{2\ln \sqrt{2}} + 1}{e^{2\ln \sqrt{2}} - 1} = \frac{e^{\ln 2} + 1}{e^{\ln 2} - 1}$$
$$\equiv \frac{2 + 1}{2 - 1} = 3.$$

10 c)

$$\coth\left(\frac{1}{4}\ln 8\right) \equiv \frac{e^{2\times\frac{1}{4}\ln 8} + 1}{e^{2\times\frac{1}{4}\ln 8} - 1}$$
$$\equiv \frac{e^{\frac{1}{2}\ln 8} + 1}{e^{\frac{1}{2}\ln 8} - 1} = \frac{\sqrt{8} + 1}{\sqrt{8} - 1}$$
$$= \frac{2\sqrt{2} + 1}{2\sqrt{2} - 1} \times \frac{2\sqrt{2} + 1}{2\sqrt{2} + 1}$$
$$= \frac{9 + 4\sqrt{2}}{7}.$$

11)

$$\coth(-x) \equiv \frac{e^{2(-x)} + 1}{e^{2(-x)} - 1}$$
$$\equiv \frac{e^{-2x} + 1}{e^{-2x} - 1} = \frac{1 + e^{2x}}{1 - e^{2x}}$$
$$\equiv \frac{e^{2x} + 1}{-(e^{2x} - 1)}$$
$$\equiv -\frac{e^{2x} + 1}{e^{2x} - 1}$$
$$\equiv -\coth x.$$

12 a)

$$\begin{aligned}\coth(-\ln 3) &\equiv -\coth(\ln 3)\\ &\equiv -\frac{e^{2\ln 3}+1}{e^{2\ln 3}-1} = -\frac{e^{\ln 9}+1}{e^{\ln 9}-1}\\ &\equiv -\frac{9+1}{9-1} = -\frac{5}{4}.\end{aligned}$$

12 b)

$$\begin{aligned}\coth(-5\ln 2) &\equiv -\coth(5\ln 2)\\ &\equiv -\frac{e^{10\ln 2}+1}{e^{10\ln 2}-1} = -\frac{e^{\ln 1024}+1}{e^{\ln 1024}-1}\\ &\equiv -\frac{1024+1}{1024-1} = -\frac{1025}{1023}.\end{aligned}$$

13 a)

$$\begin{aligned}\operatorname{cosech} x = 1 &\Rightarrow \sinh x = 1\\ &\Rightarrow x = \ln(1+\sqrt{1^2+1}) = \ln(1+\sqrt{2}).\end{aligned}$$

13 b)

$$\begin{aligned}\operatorname{cosech} x = 2 &\Rightarrow \sinh x = \frac{1}{2}\\ &\Rightarrow x = \ln\left(\frac{1}{2}+\sqrt{(\frac{1}{2})^2+1}\right)\\ &= \ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right) = \ln\left(\frac{1+\sqrt{5}}{2}\right).\end{aligned}$$

13 c)

$$\begin{aligned}\operatorname{cosech} x = -\frac{1}{5} &\Rightarrow \sinh x = -5\\ &\Rightarrow x = \ln\left(-5+\sqrt{(-5)^2+1}\right) = \ln(-5+\sqrt{26}).\end{aligned}$$

14 a)

$$\begin{aligned}\operatorname{sech} x = 1 &\Rightarrow \cosh x = 1\\ &\Rightarrow x = \ln(1+\sqrt{1^2-1}) = \ln 1 = 0.\end{aligned}$$

14 b)

$$\begin{aligned}\operatorname{sech} x = \frac{1}{2} &\Rightarrow \cosh x = 2 \\ &\Rightarrow x = \pm \ln(2 + \sqrt{2^2 - 1}) = \pm \ln(2 + \sqrt{3}) \\ &= \ln(2 \pm \sqrt{3}).\end{aligned}$$

14 c)

$$\begin{aligned}\operatorname{sech} x = \frac{2}{3} &\Rightarrow \cosh x = \frac{3}{2} \\ &\Rightarrow x = \pm \ln\left(\frac{3}{2} + \sqrt{(\frac{3}{2})^2 - 1}\right) \\ &= \pm \ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right) = \pm \ln\left(\frac{3 + \sqrt{5}}{2}\right) \\ &= \ln\left(\frac{3 \pm \sqrt{5}}{2}\right).\end{aligned}$$

15 a)

$$\begin{aligned}\coth x = 2 &\Rightarrow \tanh x = \frac{1}{2} \\ &\Rightarrow x = \frac{1}{2} \ln\left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}}\right) = \frac{1}{2} \ln 3.\end{aligned}$$

15 b)

$$\begin{aligned}\coth x = \frac{3}{2} &\Rightarrow \tanh x = \frac{2}{3} \\ &\Rightarrow x = \frac{1}{2} \ln\left(\frac{1 + \frac{2}{3}}{1 - \frac{2}{3}}\right) = \frac{1}{2} \ln 5.\end{aligned}$$

15 c)

$$\begin{aligned}\coth x = -\frac{5}{3} &\Rightarrow \tanh x = -\frac{3}{5} \\ &\Rightarrow x = \frac{1}{2} \ln\left(\frac{1 - \frac{3}{5}}{1 + \frac{3}{5}}\right) = \frac{1}{2} \ln \frac{1}{4} \\ &= -\frac{1}{2} \ln 4 = -\ln 2.\end{aligned}$$

16 a)

$$
\begin{aligned}
5\operatorname{sech} x - 2\tanh x &= 5 \\
\Rightarrow \frac{5}{\cosh x} - \frac{2\sinh x}{\cosh x} &= 5 \\
\Rightarrow 5 - 2\sinh x &= 5\cosh x \\
\Rightarrow 2\sinh x + 5\cosh x &= 5 \\
\Rightarrow 2\frac{e^x - e^{-x}}{2} + 5\frac{e^x + e^{-x}}{2} &= 5 \\
\Rightarrow 2(e^x - e^{-x}) + 5(e^x + e^{-x}) &= 10 \\
\Rightarrow 7e^x + 3e^{-x} &= 10 \\
\Rightarrow 7e^{2x} - 10e^x + 3 &= 0 \\
\Rightarrow (7e^x - 3)(e^x - 1) &= 0 \\
\Rightarrow e^x = \frac{3}{7} \text{ or } e^x &= 1 \\
\Rightarrow x = \ln\frac{3}{7} \text{ or } x &= 0.
\end{aligned}
$$

16 b)

$$
\begin{aligned}
8\operatorname{cosech} x - 3\coth x &= 27 \\
\Rightarrow \frac{8}{\sinh x} - \frac{3\cosh x}{\sinh x} &= 27 \\
\Rightarrow 27\sinh x + 3\cosh x &= 8 \\
\Rightarrow 27\frac{e^x - e^{-x}}{2} + 3\frac{e^x + e^{-x}}{2} &= 8 \\
\Rightarrow 30e^x - 24e^{-x} &= 16 \\
\Rightarrow 15e^{2x} - 8e^x - 12 &= 0 \\
\Rightarrow (3e^x + 2)(5e^x - 6) &= 0 \\
\Rightarrow e^x = -\frac{2}{3} \text{ or } e^x &= \frac{6}{5}.
\end{aligned}
$$

$e^x = -\frac{2}{3}$ has no real solutions; $e^x = \frac{6}{5} \Rightarrow x = \ln\frac{6}{5}$.

16 c)

$$
\begin{aligned}
2\tanh x - 5\coth x &= 9 \\
\Rightarrow 2\tanh x - \frac{5}{\tanh x} &= 9 \\
\Rightarrow 2\tanh^2 x - 9\tanh x - 5 &= 0 \\
\Rightarrow (2\tanh x + 1)(\tanh x - 5) &= 0 \\
\Rightarrow \tanh x = -\frac{1}{2} \text{ or } \tanh x &= 5.
\end{aligned}
$$

$\tanh x = 5$ has no real solutions. For $\tanh x = -\frac{1}{2}$,

$$x = \frac{1}{2}\ln\left(\frac{1 - \frac{1}{2}}{1 + \frac{1}{2}}\right) = \frac{1}{2}\ln\frac{1}{3}.$$

17 a)

$$\operatorname{cosech}^2 x = 9 \Rightarrow \operatorname{cosech} x = \pm 3.$$

If $\operatorname{cosech} x = 3$, then $\sinh x = \frac{1}{3}$ and

$$x = \ln\left(\frac{1}{3} + \sqrt{(\frac{1}{3})^2 + 1}\right) = \ln\left(\frac{1+\sqrt{10}}{3}\right).$$

If $\operatorname{cosech} x = -3$, then $\sinh x = -\frac{1}{3}$ and

$$x = \ln\left(-\frac{1}{3} + \sqrt{(-\frac{1}{3})^2 + 1}\right) = \ln\left(\frac{-1+\sqrt{10}}{3}\right).$$

So $x = \ln\left(\frac{1+\sqrt{10}}{3}\right)$ or $x = \ln\left(\frac{-1+\sqrt{10}}{3}\right)$.

17 b)

$$\operatorname{sech}^2 x = \frac{1}{16} \Rightarrow \operatorname{sech} x = \pm\frac{1}{4}.$$

If $\operatorname{sech} x = \frac{1}{4}$, then $\cosh x = 4$ and

$$x = \ln(4 + \sqrt{4^2 - 1}) = \ln(4 + \sqrt{15}).$$

If $\operatorname{sech} x = -\frac{1}{4}$, then $\cosh x = -4$ and the source gives

$$x = \ln(-4 + \sqrt{(-4)^2 - 1}) = \ln(-4 + \sqrt{15}).$$

So the source states $x = \ln(4 + \sqrt{15})$ or $x = \ln(-4 + \sqrt{15})$.

17 c)

$$\coth^2 x = 3 \Rightarrow \coth x = \pm\sqrt{3}.$$

If $\coth x = \sqrt{3}$, then $\tanh x = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$ and

$$x = \frac{1}{2}\ln\left(\frac{3+\sqrt{3}}{3-\sqrt{3}}\right) = \frac{1}{2}\ln(2+\sqrt{3}).$$

If $\coth x = -\sqrt{3}$, then $\tanh x = -\frac{\sqrt{3}}{3}$ and

$$x = \frac{1}{2}\ln\left(\frac{3-\sqrt{3}}{3+\sqrt{3}}\right) = \frac{1}{2}\ln(2-\sqrt{3}).$$

So $x = \frac{1}{2}\ln(2+\sqrt{3})$ or $x = \frac{1}{2}\ln(2-\sqrt{3})$.

18 a)

$$\begin{aligned}&\operatorname{cosech}^2 x + \operatorname{cosech} x - 20 = 0\\ \Rightarrow\ &(\operatorname{cosech} x + 5)(\operatorname{cosech} x - 4) = 0\\ &\Rightarrow \operatorname{cosech} x = -5 \text{ or } \operatorname{cosech} x = 4.\end{aligned}$$

If $\operatorname{cosech} x = -5$, $\sinh x = -\frac{1}{5}$, so $x = \ln\left(\frac{-1+\sqrt{26}}{5}\right)$. If $\operatorname{cosech} x = 4$, $\sinh x = \frac{1}{4}$, so $x = \ln\left(\frac{1+\sqrt{17}}{4}\right)$. Thus

$$x = \ln\left(\frac{-1+\sqrt{26}}{5}\right) \text{ or } x = \ln\left(\frac{1+\sqrt{17}}{4}\right).$$

18 b)

$$\begin{aligned}&10\operatorname{sech}^2 x - 9\operatorname{sech} x + 2 = 0\\ \Rightarrow\ &(5\operatorname{sech} x - 2)(2\operatorname{sech} x - 1) = 0\\ &\Rightarrow \operatorname{sech} x = \frac{2}{5} \text{ or } \operatorname{sech} x = \frac{1}{2}.\end{aligned}$$

If $\operatorname{sech} x = \frac{2}{5}$, then $\cosh x = \frac{5}{2}$ and $x = \ln\left(\frac{5\pm\sqrt{21}}{2}\right)$. If $\operatorname{sech} x = \frac{1}{2}$, then $\cosh x = 2$ and $x = \ln(2\pm\sqrt{3})$. So $x = \ln\left(\frac{5\pm\sqrt{21}}{2}\right)$ or $x = \ln(2 \pm \sqrt{3})$.

18 c)

$$\begin{aligned}&3\coth^2 x - 16\coth x - 64 = 0\\ \Rightarrow\ &(3\coth x + 8)(\coth x - 8) = 0\\ &\Rightarrow \coth x = -\frac{8}{3} \text{ or } \coth x = 8.\end{aligned}$$

If $\coth x = -\frac{8}{3}$, then $\tanh x = -\frac{3}{8}$ and $x = \frac{1}{2}\ln\frac{5}{11}$. If $\coth x = 8$, then $\tanh x = \frac{1}{8}$ and $x = \frac{1}{2}\ln\frac{9}{7}$. So $x = \frac{1}{2}\ln\frac{5}{11}$ or $x = \frac{1}{2}\ln\frac{9}{7}$.

19 a)

$y = \operatorname{cosech} x$ is one-to-one.

19 b)

$y = \operatorname{sech} x$ is many-to-one.

19 c)

$y = \coth x$ is one-to-one.

20 a)

$y = \operatorname{cosech} x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y \neq 0\}$.

20 b)

$y = \operatorname{sech} x$. Domain: $\{x : x \in \mathbb{R}\}$. Range: $\{y : y \in \mathbb{R},\ 0 < y \leq 1\}$.

20 c)

$y = \coth x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y < -1\} \cup \{y : y \in \mathbb{R},\ y > 1\}$.

21 a)

$y = \operatorname{cosech} x$ where the domain is $\{x : x \in \mathbb{R},\ x > 0\}$. Range: $\{y : y \in \mathbb{R},\ y > 0\}$.

21 b)

$y = \operatorname{sech} x$ where the domain is $\{x : x \in \mathbb{R},\ x > 0\}$. Range: $\{y : y \in \mathbb{R},\ 0 < y < 1\}$.

21 c)

$y = \coth x$ where the domain is $\{x : x \in \mathbb{R},\ x > 0\}$. Range: $\{y : y \in \mathbb{R},\ y > 1\}$.

22 a)

Domain: $\{x : x \in \mathbb{R},\ x \neq -3\}$. Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \neq 0\}$.

22 b)

Domain: $\{x : x \in \mathbb{R}\}$. Range: $\{g(x) : g(x) \in \mathbb{R},\ 2 < g(x) \leq 3\}$.

22 c)

Domain: $\{x : x \in \mathbb{R},\ x \neq 1\}$. Range: $\{h(x) : h(x) \in \mathbb{R},\ h(x) < -2\} \cup \{h(x) : h(x) \in \mathbb{R},\ h(x) > 0\}$.

23 a)

$f(x) = \operatorname{cosech}(x - 2) + 2$. Domain: $\{x : x \in \mathbb{R},\ x \neq 2\}$. Range: $\{f(x) : f(x) \in \mathbb{R},\ f(x) \neq 2\}$.

23 b)

$y = 2 - \operatorname{cosech} x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y \neq 2\}$.

23 c)

$g(x) = 3\operatorname{sech}(x + 3)$. Domain: $\{x : x \in \mathbb{R}\}$. Range: $\{g(x) : g(x) \in \mathbb{R},\ 0 < g(x) \leq 3\}$.

23 d)

$y = 4 + 3\operatorname{sech} x$. Domain: $\{x : x \in \mathbb{R}\}$. Range: $\{y : y \in \mathbb{R},\ 4 < y \leq 7\}$.

23 e)

$h(x) = \coth(x + 4) - 3$. Domain: $\{x : x \in \mathbb{R},\ x \neq -4\}$. Range: $\{h(x) : h(x) \in \mathbb{R},\ h(x) < -4\} \cup \{h(x) : h(x) \in \mathbb{R},\ h(x) > -2\}$.

23 f)

$y = 2 - \frac{1}{3}\coth x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y < \frac{5}{3}\} \cup \{y : y \in \mathbb{R},\ y > \frac{7}{3}\}$.

24 a)

$y = \operatorname{cosech}^2 x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y > 0\}$.

24 b)

$y = \operatorname{sech}^2 x$. Domain: $\{x : x \in \mathbb{R}\}$. Range: $\{y : y \in \mathbb{R},\ 0 < y \leq 1\}$.

24 c)

$y = \coth^2 x$. Domain: $\{x : x \in \mathbb{R},\ x \neq 0\}$. Range: $\{y : y \in \mathbb{R},\ y > 1\}$.

25 a)

$$\begin{aligned}
\operatorname{cosech} x + \operatorname{sech} x &\equiv \frac{2}{e^x - e^{-x}} + \frac{2}{e^x + e^{-x}} \\
&\equiv \frac{2e^x + 2e^{-x} + 2e^x - 2e^{-x}}{e^{2x} - e^{-2x}} \\
&\equiv \frac{4e^x}{e^{2x} - e^{-2x}} \\
&\equiv \frac{4e^{3x}}{e^{4x} - 1}.
\end{aligned}$$

25 b)

For all $x > 0$, $e^{3x} > 1$, so

$$\frac{4e^{3x}}{e^{4x} - 1} > \frac{4}{e^{4x} - 1}.$$

Therefore $\operatorname{cosech} x + \operatorname{sech} x > \frac{4}{e^{4x} - 1}$.

26)

$$\begin{aligned}
\coth x - \tanh x &\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x} \\
&\equiv \frac{e^x + e^{-x}}{e^x - e^{-x}} - \frac{e^x - e^{-x}}{e^x + e^{-x}} \\
&\equiv \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x - e^{-x})(e^x + e^{-x})} \\
&\equiv \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{e^{2x} - e^{-2x}} \\
&\equiv \frac{4}{e^{2x} - e^{-2x}} = \frac{4e^{2x}}{e^{4x} - 1}.
\end{aligned}$$

For $x > 0$, $e^{4x} > 1$, so $\frac{4e^{2x}}{e^{4x} - 1} > \frac{4e^{2x}}{e^{4x}} = 4e^{-2x}$. Hence $\coth x - \tanh x > 4e^{-2x}$.

27)

$$\begin{aligned}
f(x) &= \operatorname{cosech} x + 8 \operatorname{sech} x, \\
f'(x) &= -\operatorname{cosech} x \coth x - 8 \operatorname{sech} x \tanh x
\end{aligned}$$

$$= -\frac{\cosh x}{\sinh^2 x} - \frac{8\sinh x}{\cosh^2 x}$$
$$= \frac{-\cosh^3 x - 8\sinh^3 x}{\sinh^2 x \cosh^2 x}.$$

Stationary points exist when $f'(x) = 0$:

$$-\cosh^3 x - 8\sinh^3 x = 0$$
$$-1 - 8\tanh^3 x = 0$$
$$\tanh^3 x = -\frac{1}{8}$$
$$\tanh x = -\frac{1}{2}$$
$$x = \frac{1}{2}\ln\left(\frac{1-\frac{1}{2}}{1+\frac{1}{2}}\right) = \frac{1}{2}\ln\frac{1}{3}.$$

$$f\left(\frac{1}{2}\ln\frac{1}{3}\right) = \operatorname{cosech}\left(\frac{1}{2}\ln\frac{1}{3}\right) + 8\operatorname{sech}\left(\frac{1}{2}\ln\frac{1}{3}\right)$$
$$= \frac{2}{1/\sqrt{3} - \sqrt{3}} + \frac{16}{1/\sqrt{3} + \sqrt{3}}$$
$$= \frac{2\sqrt{3}}{1-3} + \frac{16\sqrt{3}}{1+3} = -\sqrt{3} + 4\sqrt{3} = 3\sqrt{3}.$$

Range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \leq 3\sqrt{3}\}$.

28)

$$f(x) = \cosh x + \coth x,$$
$$f'(x) = \sinh x - \operatorname{cosech}^2 x = \sinh x - \frac{1}{\sinh^2 x}$$
$$= \frac{\sinh^3 x - 1}{\sinh^2 x}.$$

Stationary points exist when $f'(x) = 0$:

$$\sinh^3 x - 1 = 0$$
$$\sinh x = 1$$
$$x = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2}).$$

$$f(\ln(1+\sqrt{2})) = \cosh(\ln(1+\sqrt{2})) + \coth(\ln(1+\sqrt{2}))$$
$$= \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{2} + \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}$$
$$= \frac{1+\sqrt{2}+\sqrt{2}-1}{2} + \frac{1+\sqrt{2}+\sqrt{2}-1}{1+\sqrt{2}-(\sqrt{2}-1)}$$
$$= \sqrt{2} + \sqrt{2} = 2\sqrt{2}.$$

Range is $f(x) \in [2\sqrt{2}, \infty)$.